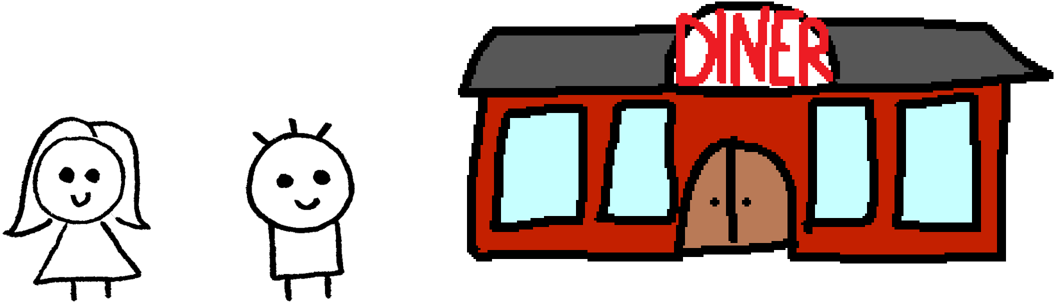
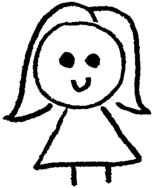
On the Expressive Power of Languages for Static Variability

P. Bittner, A. Schultheiß, B. Moosherr, J. Young, L. Teixeira, E. Walkingshaw, P. Ataei, Thomas Thüm | SE'26 | OOPSLA'24

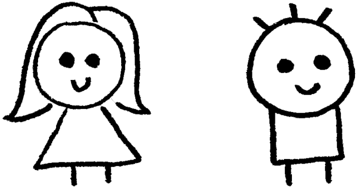
universität ulm
PADERBORN UNIVERSITY
u^b Universität Bern
INPUT | OUTPUT
Universidade Federal de Pernambuco
Technische Universität Braunschweig


DINER



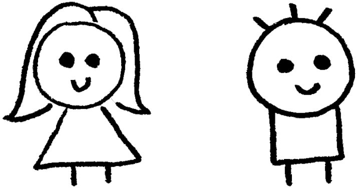

VEGETARIAN
WHICH WICH WOULD YOU LIKE?
TRIPLE CHEESE MELT
ELVIS WICH
TOMATO & AVOCADO
BLACK BEAN PATTY
HUMMUS & BELL PEPPERS
CHOOSE YOUR BREAD
WHITE
WHEAT
CHOOSE YOUR CHEESE (Optional)
AMERICAN
SWISS
PROVOLONE
CHEDDAR
PEPPER JACK
MOZZARELLA
How Would You Like Your WICH Worked?
MUSTARDS
Yellow
Dijon
Honey
Deli
MAYOS
Regular
Lite
Horseradish
Spicy
SPREADS & SAUCES
BBQ
Buffalo
Marinara
1000 Island
Ranch
ONIONS
Red
Grilled
Crispy Strings
VEGGIES
Lettuce
Tomato
Pickles
Jalapenos
Olive Salad
Mushrooms
Sauerkraut
Coleslaw
Bell Peppers
OILS & SPICES
Oil
Vinegar
Parmesan
Oregano

This eerily reminds me of work...



This eerily reminds me of work...

Indeed! The sandwich variety resembles the variability in our code base!



Inde
sandw
resen
variab
cod

```
  static void
f_foreground(/* params */)
{
#ifdef FEAT_GUI
  if (gui.in_use)
  gui_mch_set_foreground();
#else
# ifdef MSWIN
  win32_set_foreground();
# endif
#endif
}
```

Vim Commit ab4cece

Inde
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}
```

#define FEAT_GUI 0,
#define MSWIN 1

```c
   static void
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}
```

Vim Commit ab4cece

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sandw
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#define FEAT_GUI 0,
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```c
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}
```

#define FEAT_GUI 0,
#define MSWIN 0 →

```c
static void
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{
}
```
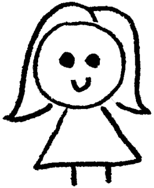
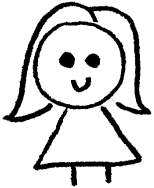

Vim Commit ab4cece

And its not just our code and sandwiches, I guess...



True, there is variability in many domains!



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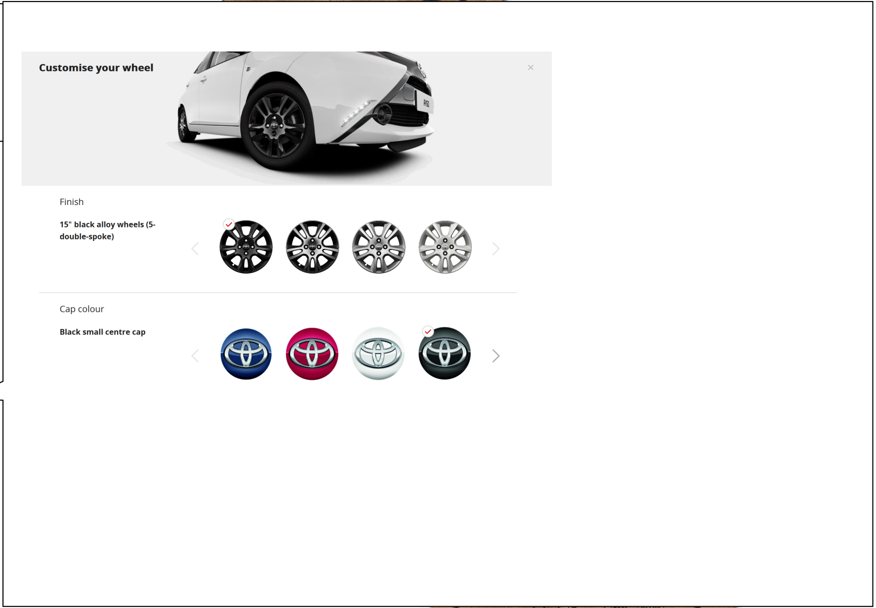

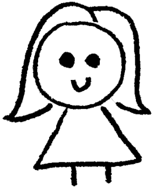

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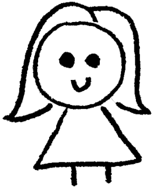

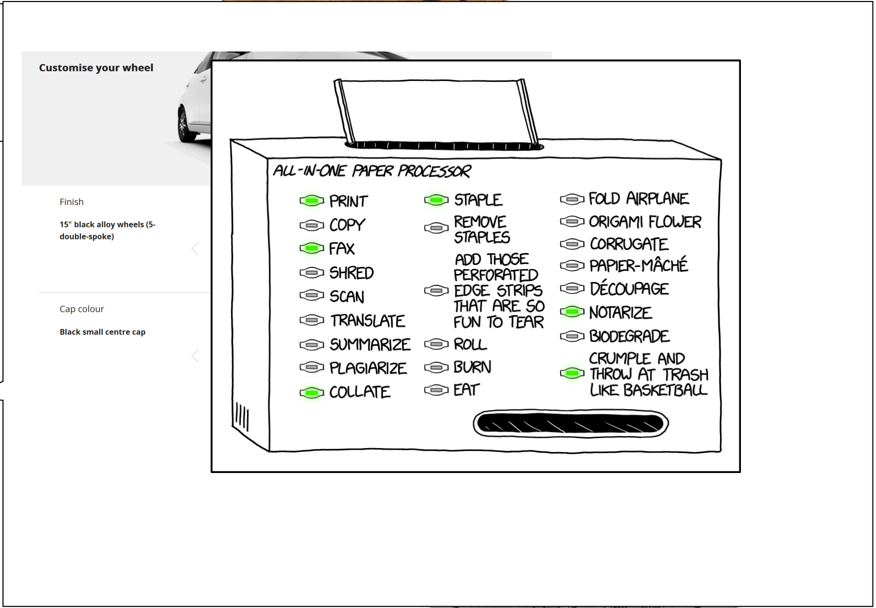

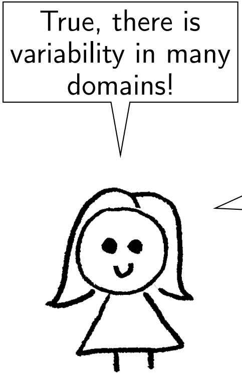
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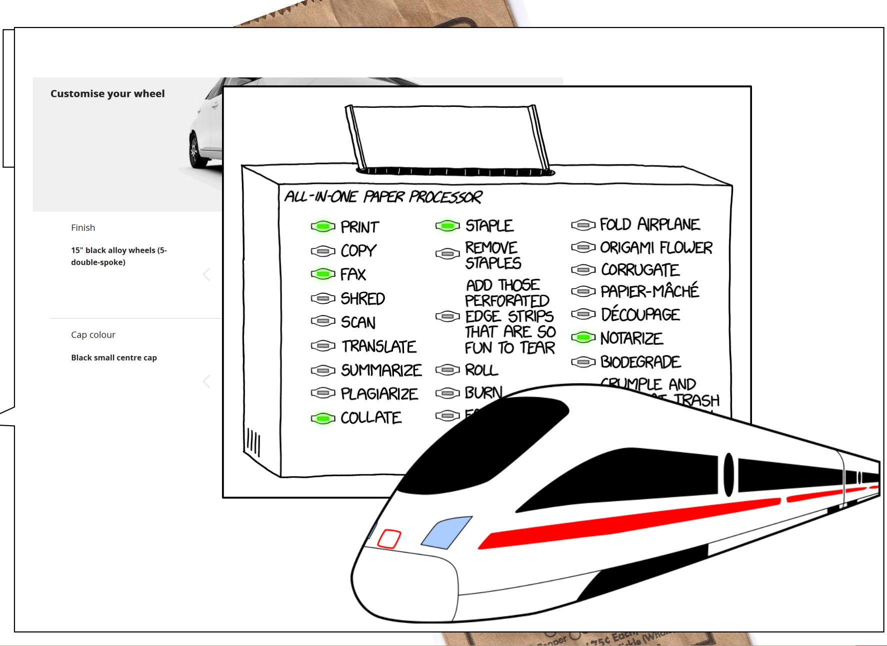
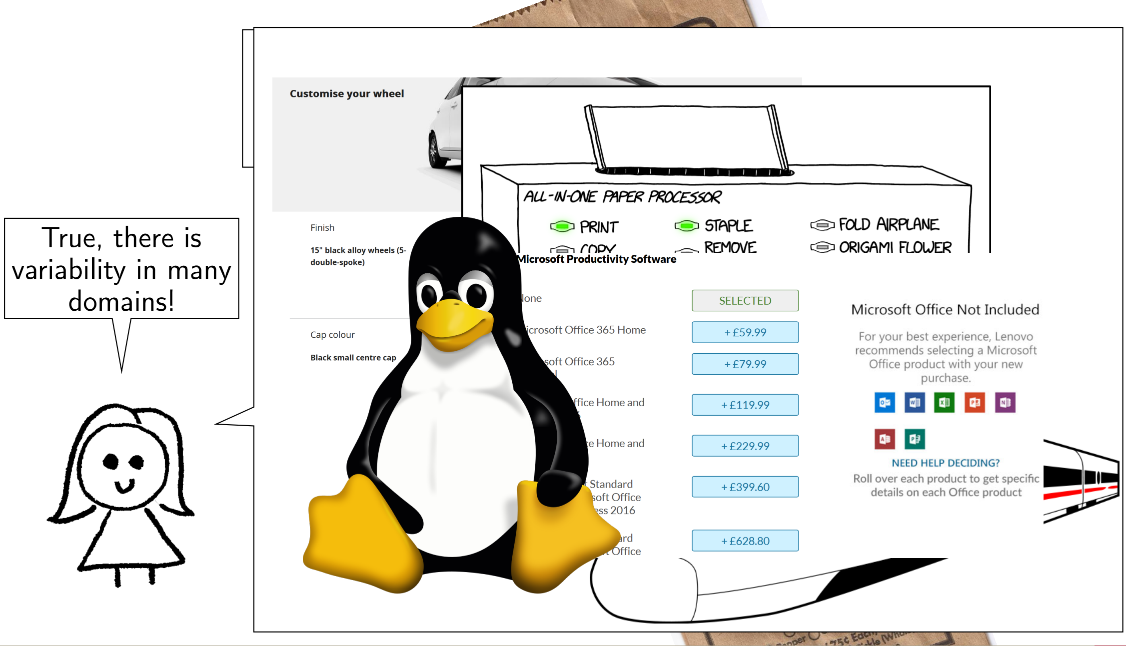

Customise your wheel
Finish
15" black alloy wheels (5-double-spoke)
Cap colour
Black small centre cap
ALL-IN-ONE PAPER PROCESSOR
PRINT
COPY
FAX
SHRED
SCAN
TRANSLATE
SUMMARIZE
PLAGIARIZE
COLLATE
STAPLE
REMOVE STAPLES
ADD THOSE PERFORATED EDGE STRIPS THAT ARE SO FUN TO TEAR
ROLL
BURN
FOLD AIRPLANE
ORIGAMI FLOWER
CORRUGATE
PAPIER-MÂCHÉ
DÉCOUPAGE
NOTARIZE
BIODEGRADE


True, there is variability in many domains!
Customise your wheel
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Cap colour
Black small centre cap
ALL-IN-ONE PAPER PROCESSOR
PRINT
STAPLE
FOLD AIRPLANE
REMOVE
ORIGAMI FLOWER
Microsoft Productivity Software
SELECTED
+ £59.99
+ £79.99
+ £119.99
+ £229.99
+ £399.60
+ £628.80
Microsoft Office Not Included
For your best experience, Lenovo recommends selecting a Microsoft Office product with your new purchase.
NEED HELP DECIDING?
Roll over each product to get specific details on each Office product

But how to *describe* and *analyze* variability across domains?

	Computation	Static Variability
Implementation		
Specification		

	Computation	Static Variability
Implementation	Java, C, Haskell, OCaml, Prolog, ...	
Specification		

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Implementation	Java, C, Haskell, OCaml, Prolog, ...	
Specification	Turing Machine, Lambda Calculus, Type-0 Grammar, ...	

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	Computation	Static Variability
Implementation	Java, C, Haskell, OCaml, Prolog, ...	C Preprocessor, Mixins, Aspects, Sandwich Menus, ...
Specification	Turing Machine, Lambda Calculus, Type-0 Grammar, ...	?

Core Choice Calculus [Erwig and Walkingshaw, 2011] [Walkingshaw, 2013] – A Lambda Calculus of Variability?

$$
\begin{array}{rcll}
e & ::= & a\prec e, \ldots, e\succ & \textit{Object Structure} \\
  & \mid & D\langle e, \ldots, e\rangle & \textit{Choice}
\end{array}
$$

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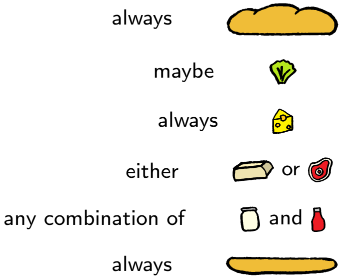

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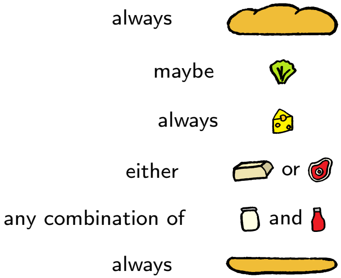

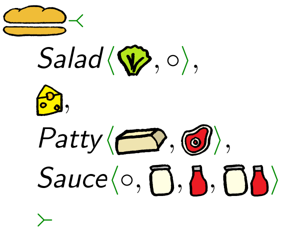

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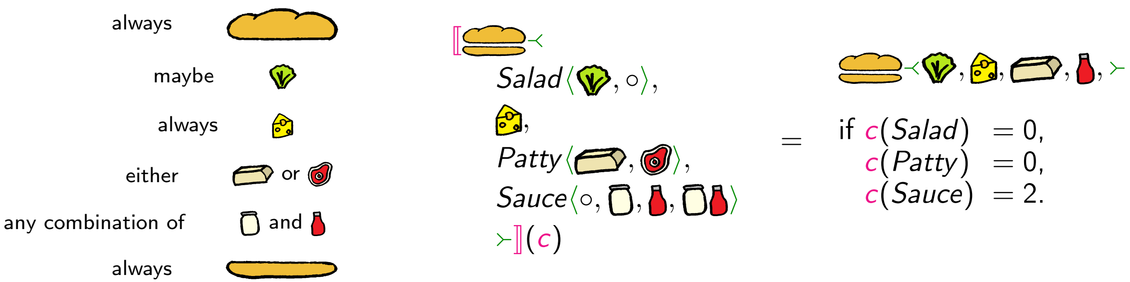
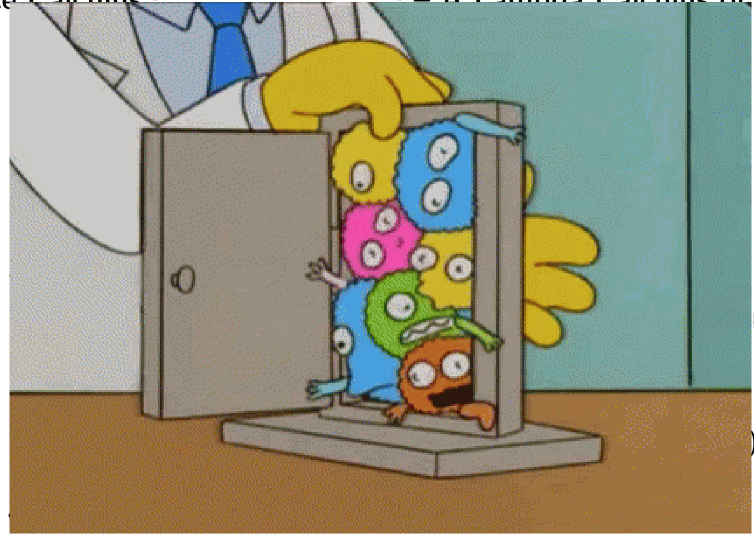


Core Choice Calculus [Erwig and Walkingshaw, 2011] – A Lambda Calculus of Variability?



always

maybe

always

either

any combination of

always

🧀, 🧈, 🥫, $\succ$

$= 0,$
$= 0,$
$) = 2.$

Choice Calculus
[Erwig and Walkingshaw, 2011]
[Walkingshaw, 2013]

Artifact Trees
[Linsbauer et al., 2017]

Feature Structure Trees
[Apel et al., 2013]

Algebraic Decision Trees
[Bahar et al., 1993]

Gruler's Language
[Gruler, 2010]

Clone and Own

Variability-Aware ASTs
[Kästner et al., 2008]

Variational IMP
[Midtgaard et al., 2015]

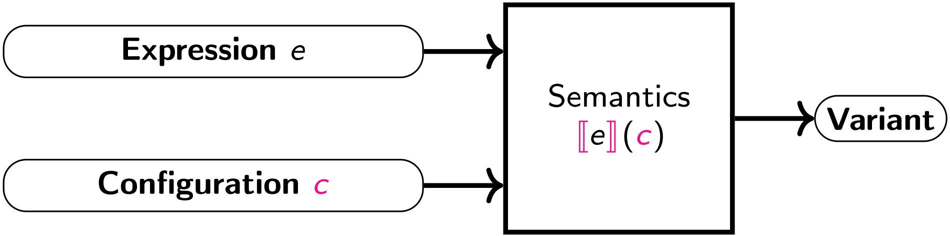
...

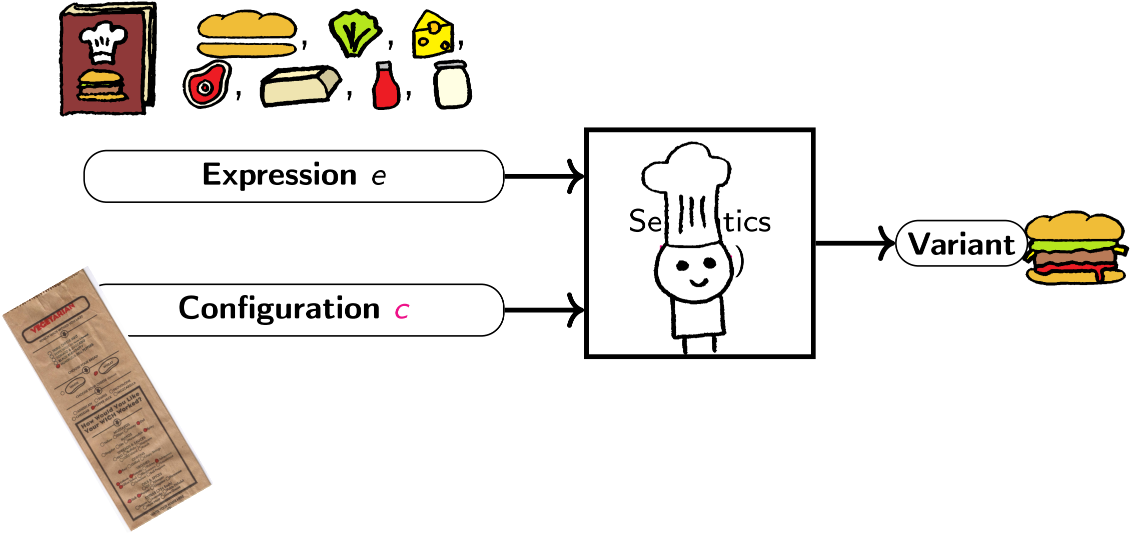
But how do these languages relate? Can we transfer research results based on one formalism to the others?

Variability-Aware ASTs
[Kästner et al., 2008]

Variational IMP
[Midtgaard

...


Expression e
Configuration c
Semantics
$[\![e]\!](c)$
Variant


Expression e
Configuration c
Se tics
Variant
VEGETARIAN
How Would You Like Your WICH Worked?

How to Compare Variability Languages L and M semantically?

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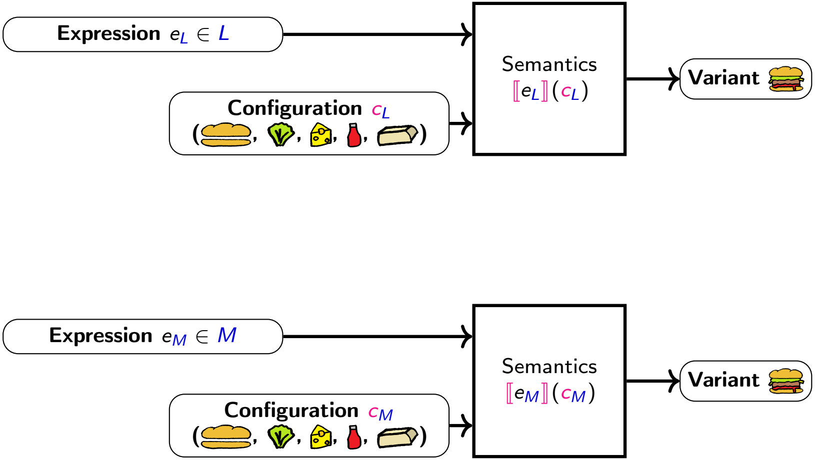
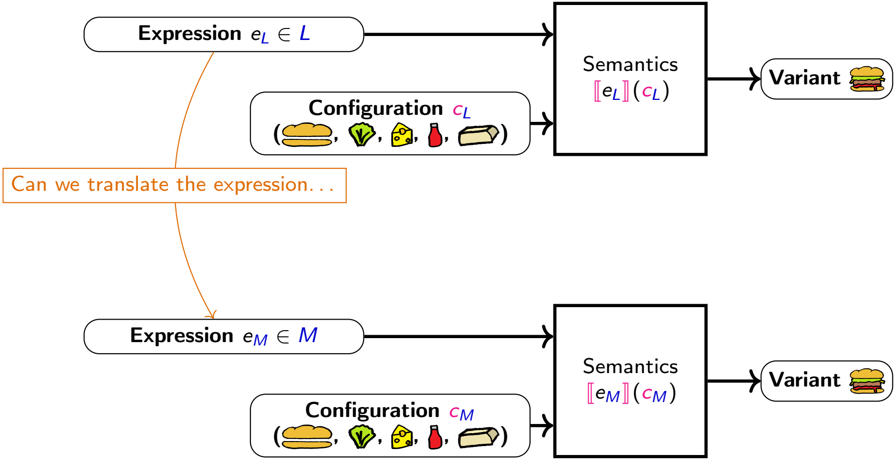


How to Compare Variability Languages L and M semantically?



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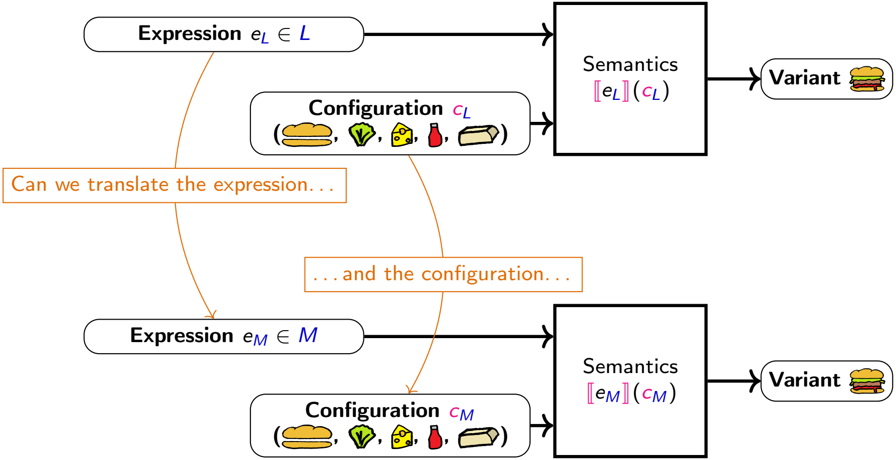

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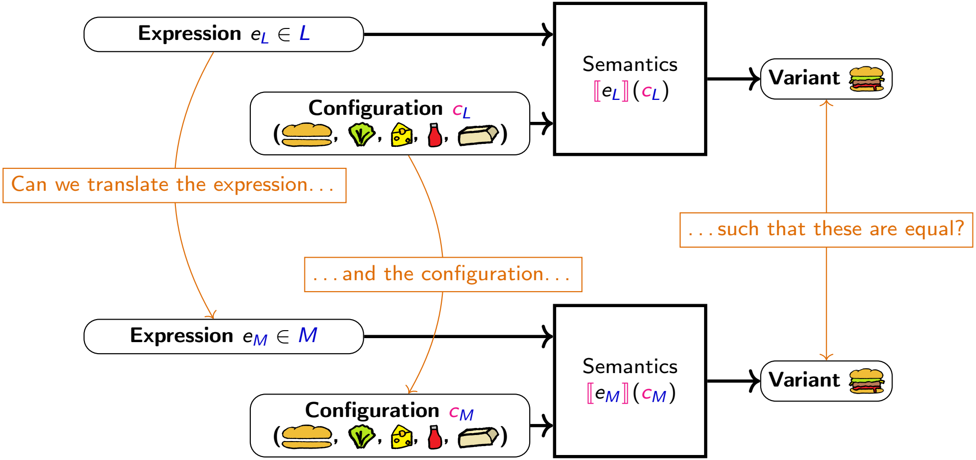

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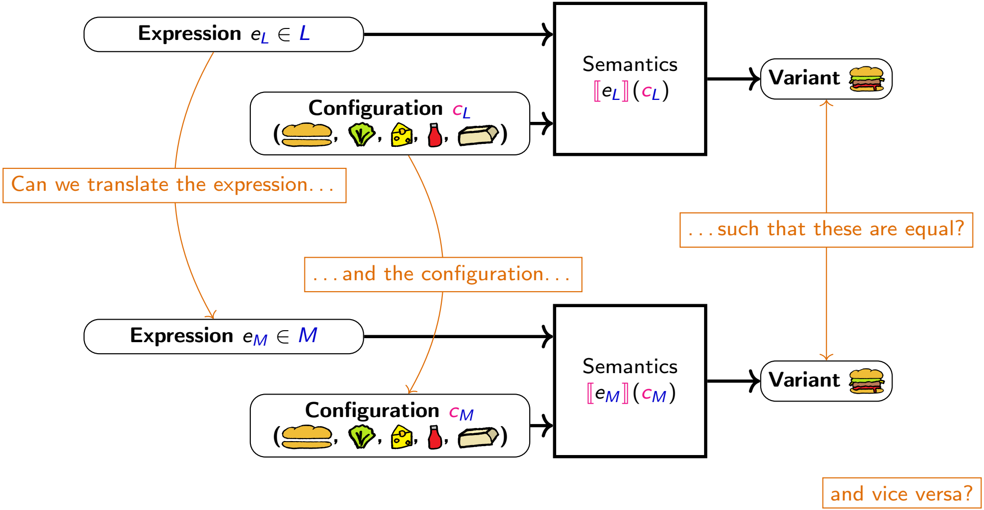
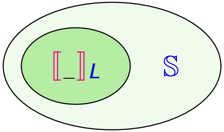
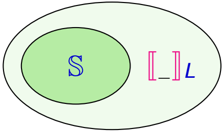


$\mathrm{Sound}(\mathbb{S}, L) \quad :=$



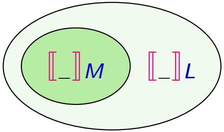
(Upper bound for expressiveness)

$\mathrm{Complete}(\mathbb{S}, L) \quad :=$



(Lower bound for expressiveness)

$L \succeq M \quad :=$



(Relative expressiveness)

Binary Choice
Calculus

Option
Calculus

Algebraic
Decision Trees

Core Choice
Calculus

Feature
Structure Trees

n-ary Choice Calculus

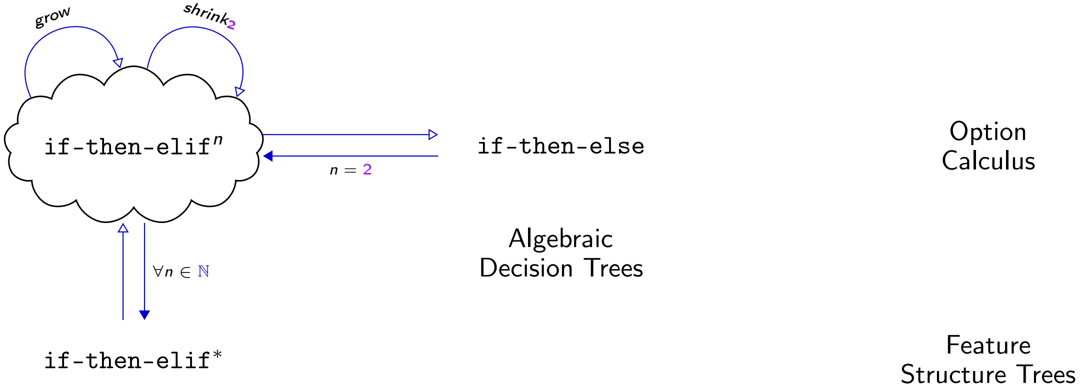
Binary Choice Calculus

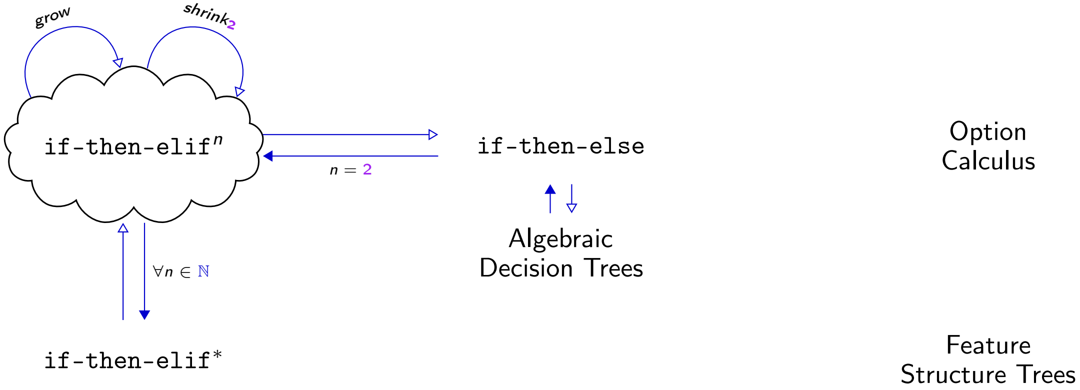
Option Calculus

Algebraic Decision Trees

Core Choice Calculus

Feature Structure Trees


grow
shrink$_2$
if-then-elif^n
if-then-else
$n = 2$
$\forall n \in \mathbb{N}$
if-then-elif*
Algebraic Decision Trees
Option Calculus
Feature Structure Trees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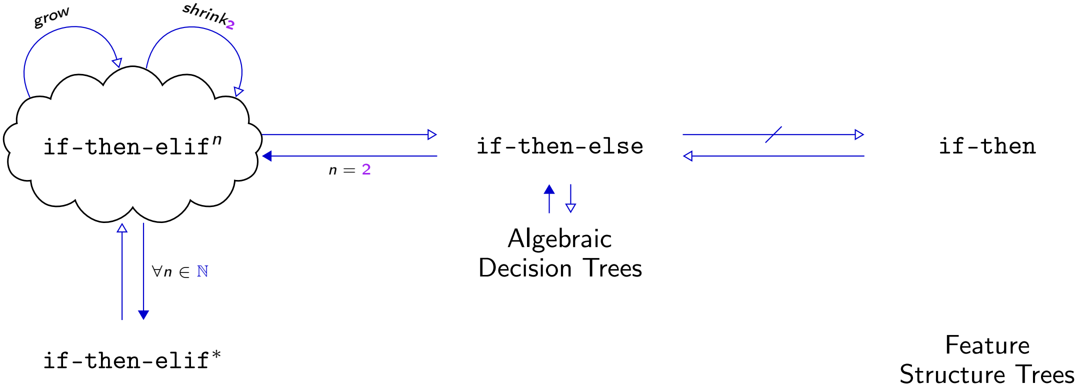
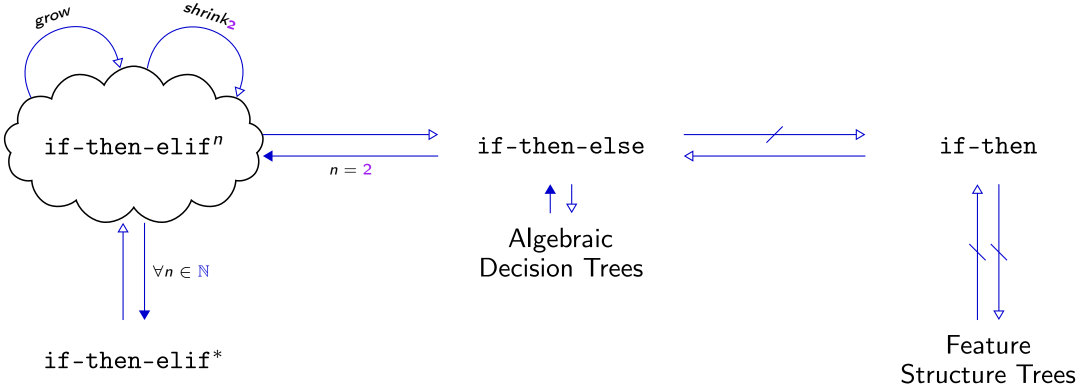

grow
shrink2
if-then-elif^n
n = 2
if-then-else
if-then
∀n ∈ ℕ
Algebraic Decision Trees
if-then-elif*
Feature Structure Trees


grow
shrink₂
if-then-elif^n
$n = 2$
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if-then
$\forall n \in \mathbb{N}$
Algebraic
Decision Trees
if-then-elif*
Feature
Structure Trees

grow

shrink$_2$

Binary Choice Calculus

Option Calculus

Algebraic Decision Trees

Clone and Own

Feature Structure Trees

grow shrink$_2$

Binary Choice Calculus

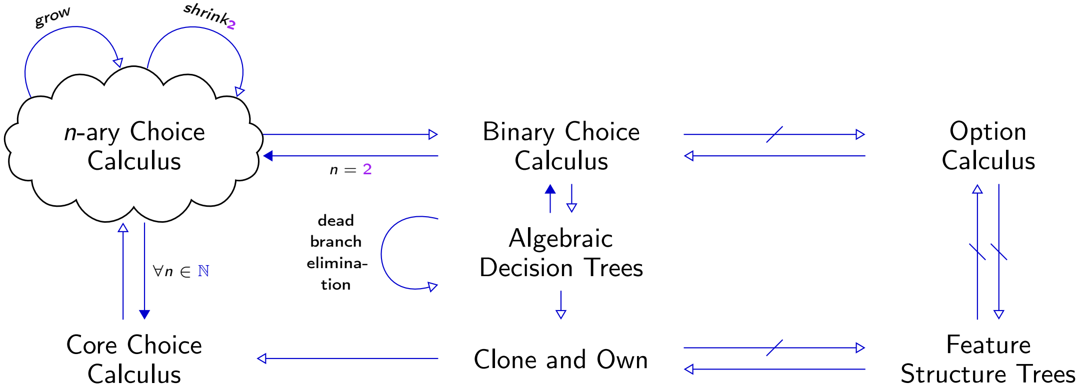
Option Calculus

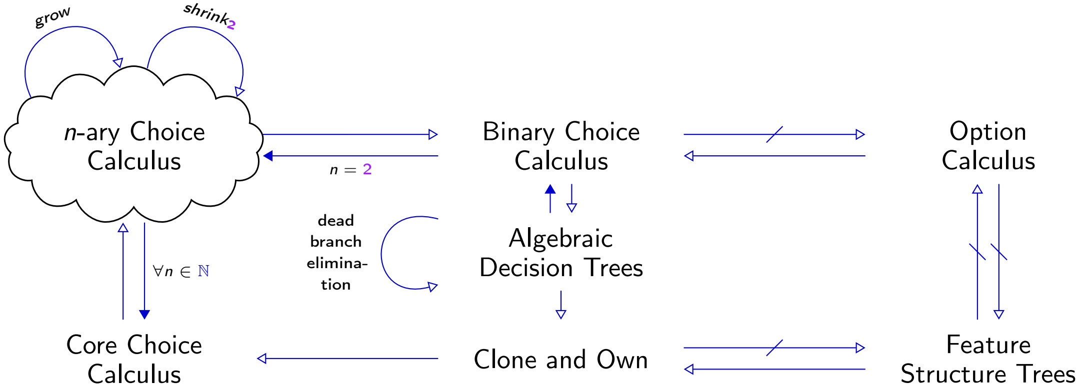
Algebraic Decision Trees

Clone and Own

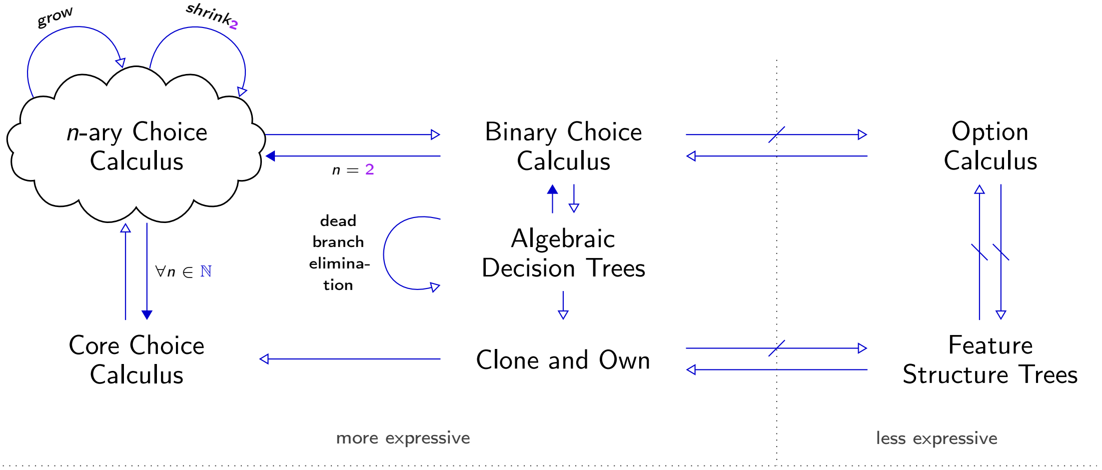
Feature Structure Trees

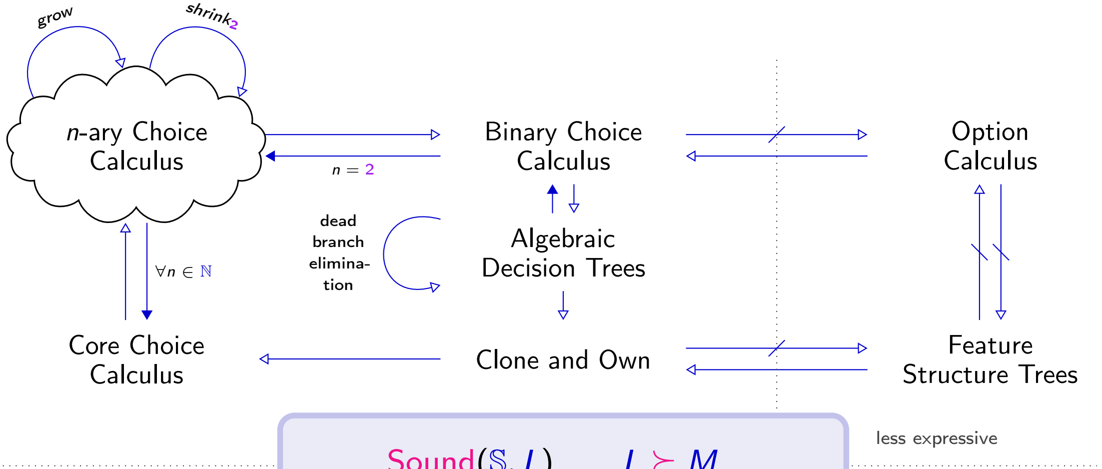
We prove this sound and complete on finite, non-empty sets of variants.


grow
shrink₂
n-ary Choice Calculus
Binary Choice Calculus
Option Calculus
n = 2
dead branch elimination
Algebraic Decision Trees
∀n ∈ ℕ
Core Choice Calculus
Clone and Own
Feature Structure Trees

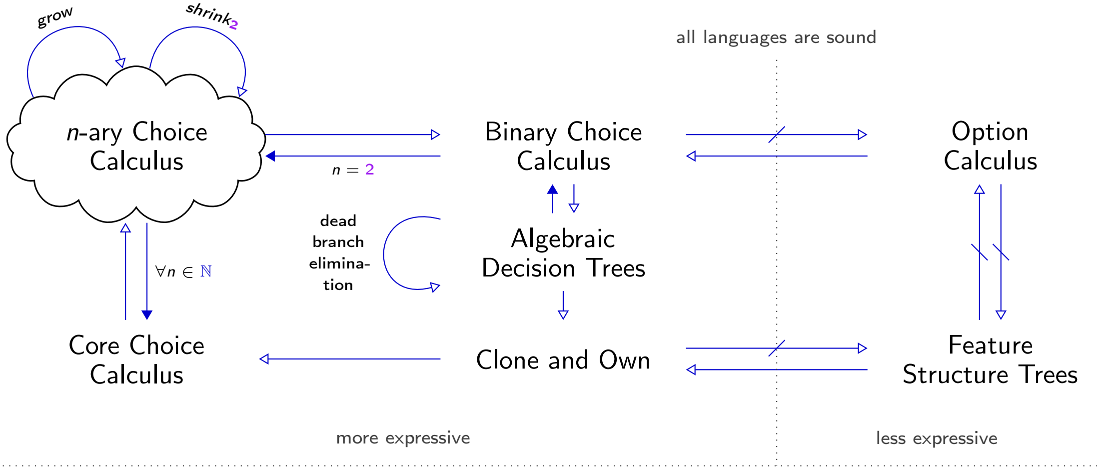


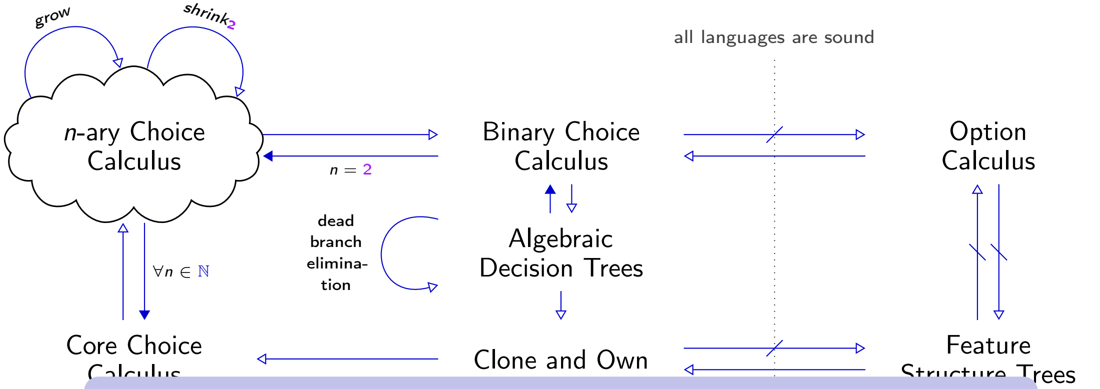
$$\frac{\text{Complete}(\mathbb{S}, L) \qquad M \succeq L}{\text{Complete}(\mathbb{S}, M)}$$


grow
shrink2
n-ary Choice Calculus
n = 2
Binary Choice Calculus
Option Calculus
dead branch elimina- tion
Algebraic Decision Trees
∀n ∈ ℕ
Core Choice Calculus
Clone and Own
Feature Structure Trees
more expressive
less expressive



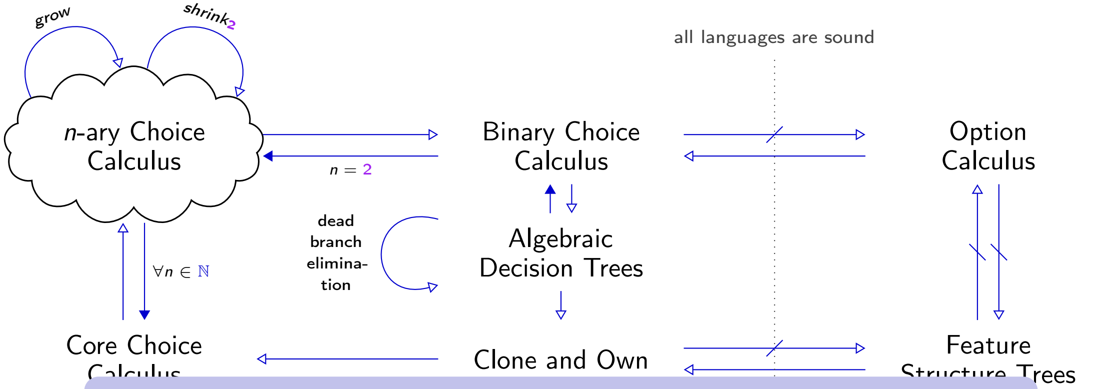
$$\frac{\text{Sound}(\mathbb{S}, L) \qquad L \succeq M}{\text{Sound}(\mathbb{S}, M)}$$


grow
shrink2
n-ary Choice Calculus
Binary Choice Calculus
Option Calculus
n = 2
all languages are sound
dead branch elimination
Algebraic Decision Trees
∀n ∈ ℕ
Core Choice Calculus
Clone and Own
Feature Structure Trees
more expressive
less expressive



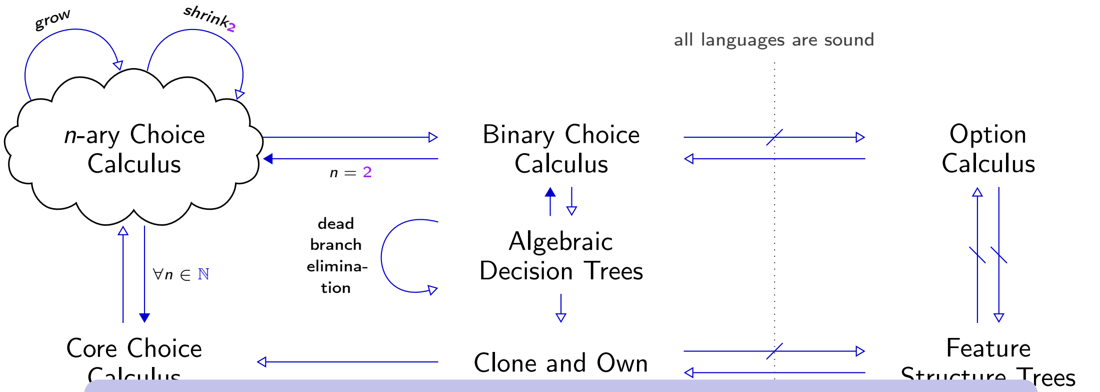
Conclusions

- Choice calculus is indeed a lambda calculus of variation



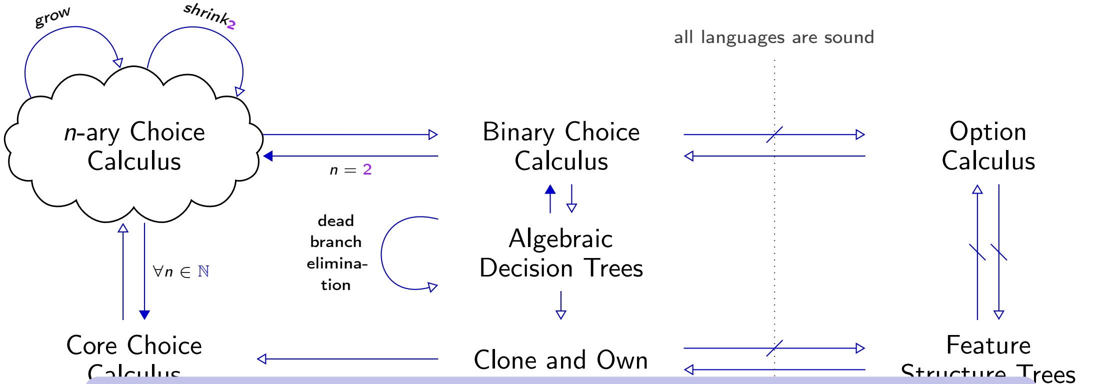
Conclusions

- Choice calculus is indeed a lambda calculus of variation as well as algebraic decision trees and clone-and-own.



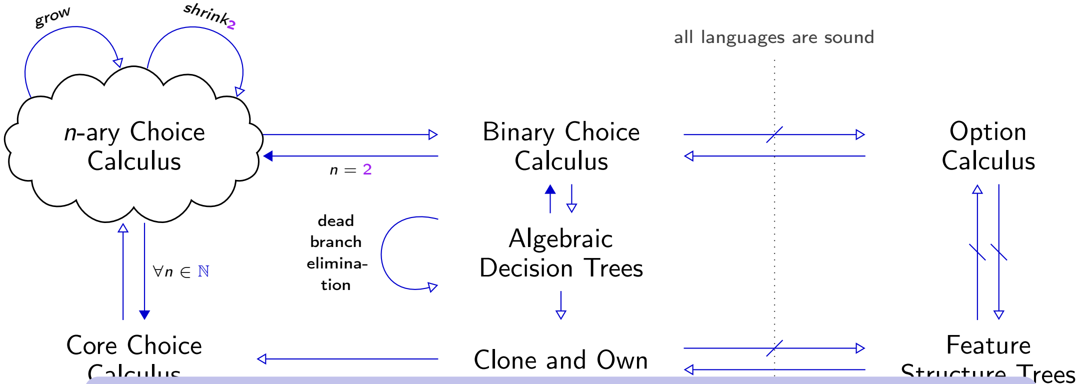
Conclusions

- Choice calculus is indeed a lambda calculus of variation as well as algebraic decision trees and clone-and-own.
- There are ≥ 3 classes of expressiveness, arising from different syntactical restrictions.



Conclusions

- Choice calculus is indeed a lambda calculus of variation as well as algebraic decision trees and clone-and-own. $\Rightarrow$ **Are all languages equally useful?**
- There are ≥ 3 classes of expressiveness, arising from different syntactical restrictions.



Conclusions

- Choice calculus is indeed a lambda calculus of variation as well as algebraic decision trees and clone-and-own. $\Rightarrow$ **Are all languages equally useful?**
- There are ≥ 3 classes of expressiveness, arising from different syntactical restrictions. $\Rightarrow$ **What is the value of less expressive languages?**

Are all maximal expressive languages equally useful?

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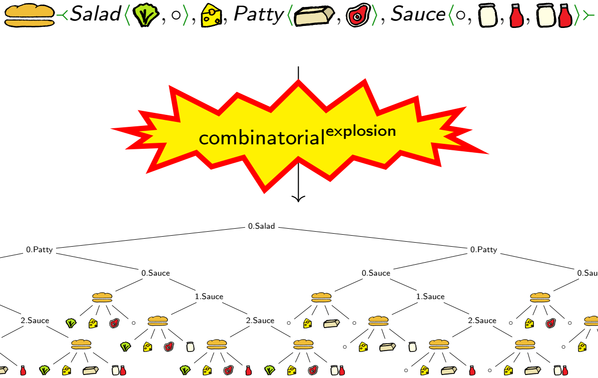

for algebraic decision trees and clone-and-own

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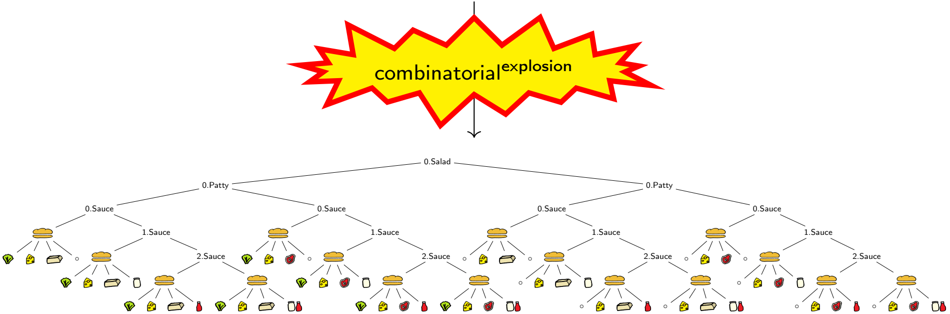

for algebraic decision trees and clone-and-own

No, avoiding duplication of shared sub-terms is essential for scalability and usability.

(known problem of clone-and-own and a basic principle of software engineering)

What is the value of less expressive languages?

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Encoding options with choices requires context-sensitive neutral values ∘

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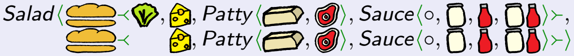

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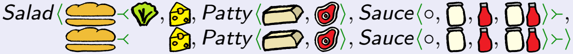

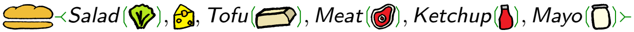

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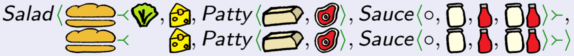

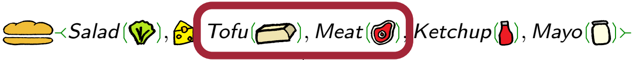

Options avoid these drawbacks but cannot express alternatives! (That is why they are incomplete.)

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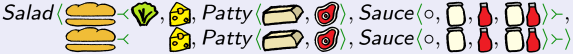

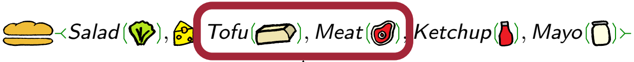

Options avoid these drawbacks but cannot express alternatives! (That is why they are incomplete.)

Conclusion: Use choices and options for completeness and usability.

Everything is formalized in Agda

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Proofs are programs:

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A **soundness** proof enumerates all variants:

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Proofs are programs:

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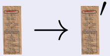
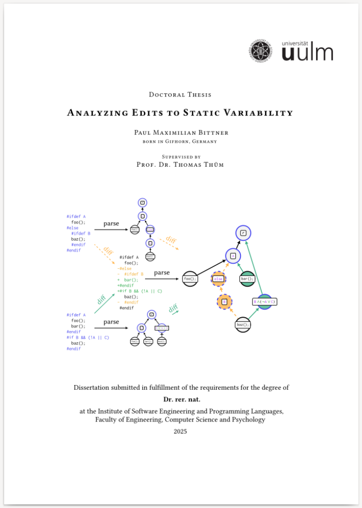
An **expressiveness** proof is a compiler:



A revised version of this work is part of Paul's PhD thesis.


universität ulm

Doctoral Thesis

Analyzing Edits to Static Variability

Paul Maximilian Bittner
born in Gifhorn, Germany

Supervised by
Prof. Dr. Thomas Thüm

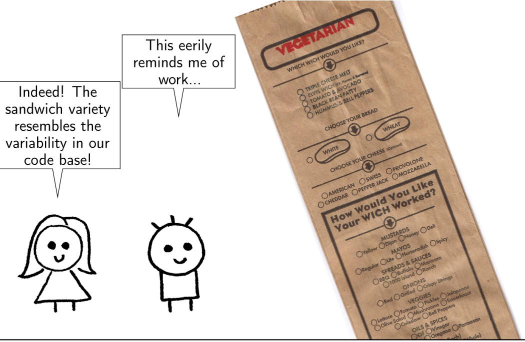
Dissertation submitted in fulfillment of the requirements for the degree of

Dr. rer. nat.

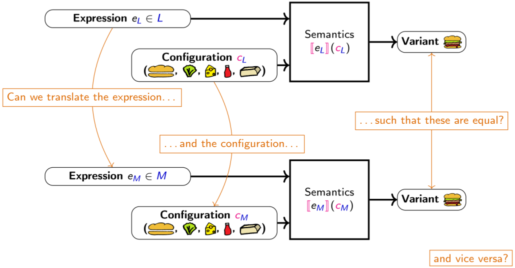
at the Institute of Software Engineering and Programming Languages,
Faculty of Engineering, Computer Science and Psychology

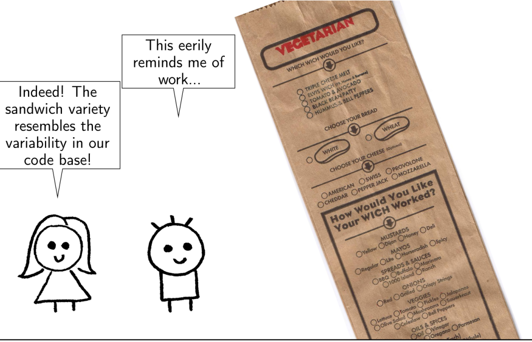
2025

DOI: 10.18725/OPARU-58798


Indeed! The sandwich variety resembles the variability in our code base!
This eerily reminds me of work...
VEGETARIAN
WHICH WICH WOULD YOU LIKE?
CHOOSE YOUR BREAD
WHITE
WHEAT
CHOOSE YOUR CHEESE
How Would You Like Your WICH Worked?
MUSTARDS
MAYOS
SPREADS & SAUCES
ONIONS
VEGGIES
OILS & SPICES

How to Compare Variability Languages L and M semantically?





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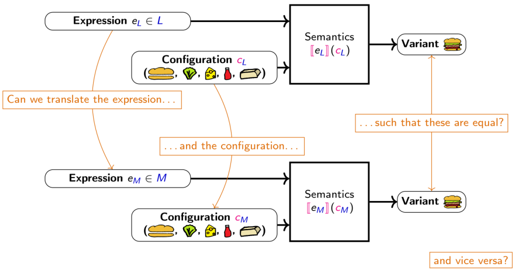

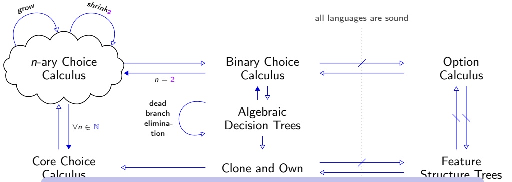

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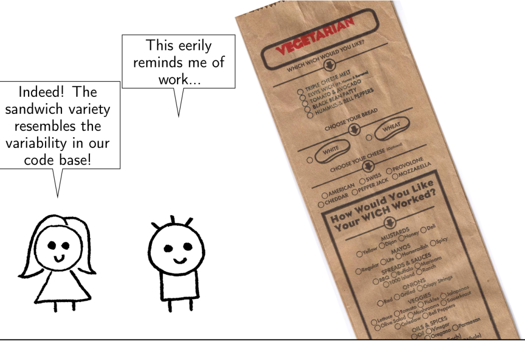

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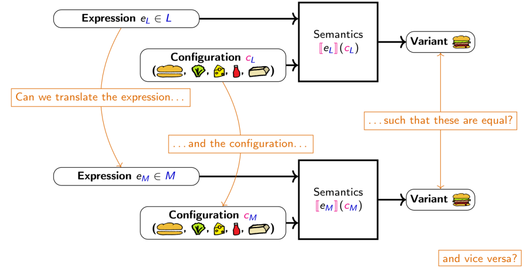

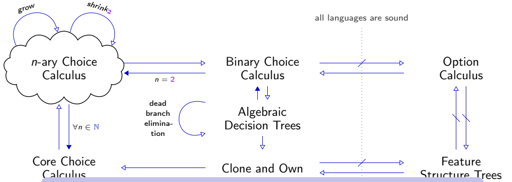

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